

RANGE OF MOTION MEASUREMENTS

GONIOMETRY

Shoulder Flexion:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with hips and knees bent and lumbar spine flat. Arm is at the side with the palm in and the thumb up
Stabilization:	Body weight should stabilize scapula but manual stabilization may be required to prevent excessive scapular rising and tipping posteriorly
Goniometer Axis:	Lateral aspect of the center of the humeral head approximately 1" below the acromion process
Stationary Arm:	Parallel to midaxillary line of the trunk
Moving Arm:	Parallel to longitudinal axis of the humerus pointing toward the lateral epicondyle
Movement:	Shoulder flexion
Expected ROM:	120° of pure GH flexion; 150° with GH, AC, SC, and scapulothoracic contribution; 180° if lumbar hyperextension permitted
Substitutions:	Lumbar hyperextension Scapular tipping Maintain slight elbow flexion so that long head of triceps does not restrict motion

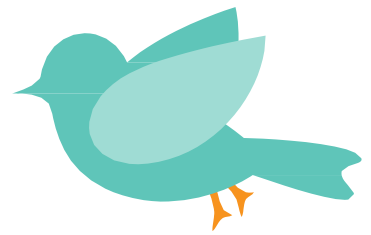
Shoulder Abduction: (note plane of movement - scapular vs. frontal plane of elevation)

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with the hips and knees bent. Arm is in the anatomical position with the shoulder externally rotated
Stabilization:	Stabilize the thorax
Goniometer Axis:	Anterior aspect of the shoulder just inferior and lateral to the coracoid process
Stationary Arm:	Parallel to midaxillary line of the trunk
Moving Arm:	Anterior aspect of the upper arm parallel to longitudinal axis of the humerus
Movement:	Shoulder elevation in scapular or frontal plane
Expected ROM:	90° of pure GH abduction; 150° with GH, AC, SC, and scapulothoracic contribution; 180° if lumbar lateral flexion is allowed
Substitutions:	Lumbar lateral flexion Excessive scapular upward rotation contribution to movement

Shoulder Internal/External Rotation

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with the shoulder and elbow abducted 90°. The forearm is midway between pronation/supination with the entire humerus is supported by the table.
Stabilization:	Stabilize the distal humerus through the full ROM and stabilize the thorax/scapula at the end ROM
Goniometer Axis:	The olecranon process of the ulna projecting through the humeral shaft toward the humeral head
Stationary Arm:	Parallel to the supporting surface or perpendicular to the floor
Moving Arm:	Parallel to the longitudinal axis of the ulna pointing toward the styloid process
Movement:	Internal and External Rotation
Expected ROM:	70° internal rotation; 90° external rotation
Substitutions:	Elbow extension, scapular elevation, tilting, or protraction The amount of motion available is influenced by the position of abduction in the frontal plane and whether the measurements are performed in the scapular or frontal planes. Specifically record the position during measurement.





Shoulder Horizontal Abduction:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with the hips and knee bent and the shoulder abducted 90° and the elbow flexed 90°
Stabilization:	Stabilize the thorax to prevent lateral lumbar flexion or trunk roll
Goniometer Axis:	Superior surface of the acromion
Stationary Arm:	Perpendicular to the floor projecting towards the floor
Moving Arm:	Parallel to the longitudinal axis of the humerus pointing toward the lateral epicondyle
Movement:	Horizontal abduction
Expected ROM:	45°
Substitutions:	Trunk Rotation

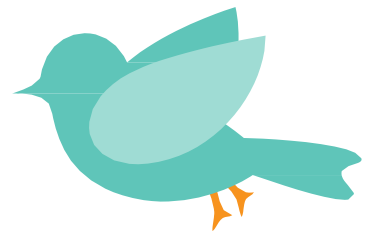
Elbow Flexion:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine or sitting with the arm parallel to the midline and the forearm in the anatomical position
Stabilization:	Examiner manually stabilizes the humerus
Goniometer Axis:	Over the lateral epicondyle of the humerus
Stationary Arm:	Parallel to the longitudinal axis of the humerus pointing towards the tip of the acromion
Moving Arm:	Parallel to longitudinal axis of the radius pointing toward the styloid process of the radius
Movement:	Elbow flexion
Expected ROM:	150°
Stabilization:	Prevent shoulder flexion
Note:	If the forearm cannot be placed in full supination, note the position of testing

Elbow Extension:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine or sitting with the arm parallel to the midline and the forearm in the anatomical position
Stabilization:	Examiner manually stabilizes the humerus
Goniometer Axis:	Over the lateral epicondyle of the humerus
Stationary Arm:	Parallel to the longitudinal axis of the humerus pointing towards the tip of the acromion
Moving Arm:	Parallel to the longitudinal axis of the radius pointing towards the styloid process of the radius
Movement:	Elbow extension
Expected ROM:	0° in males; 10-15° in females is common.
Stabilization:	Prevent scapular tilting
Note:	if hyperextension range is available, a towel roll can be placed under the humerus





Forearm Pronation/Supination:

Measurement Tool:	Universal Goniometer
Testing Position:	Sitting with the shoulder adducted to the side and the elbow flexed 90° with the forearm in the neutral "hand shake" midposition.
Stabilization:	Examiner stabilizes the humerus and does not allow abduction or adduction
Goniometer Axis:	Just medial and posterior (supination) or lateral and posterior (pronation) to the ulnar styloid process
Stationary Arm:	Parallel to the long axis of the humerus
Moving Arm:	Parallel to the dorsal forearm surface in pronation and volar surface in supination
Movement:	External motion into full supination and internal motion into full pronation
Expected ROM:	80-90° of pronation and supination
Substitutions:	<i>Supination:</i> wrist extension and/or radial deviation, adduction and ER of the shoulder, and ipsilateral trunk lateral flexion <i>Pronation:</i> wrist flexion and/or ulnar deviation, abduction and IR of the shoulder, and/or contralateral trunk lateral flexion

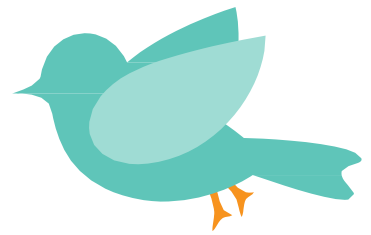
Wrist Flexion-Extension:

Measurement Tool:	Universal Goniometer
Testing Position:	Elbow flexed 90°; Wrist over edge of table with forearm in full pronation
Stabilization:	Forearm stabilized to prevent pronation or supination
Goniometer Axis:	Lateral placement over the triquetrum
Proximal Arm:	Parallel with the ulna bisecting the ulnar styloid, radial head, and lateral epicondyle
Distal Arm:	Parallel to longitudinal axis of the fifth metacarpal
Movement:	Wrist flexion with fingers extended and extension with fingers comfortably flexed
Expected ROM:	75° for both flexion and extension
Substitutions:	Excessive radial or ulnar deviation

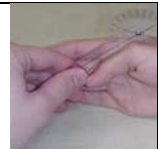
Wrist Radial-Ulnar Deviation:

Measurement Tool:	Universal Goniometer
Testing Position:	Elbow flexed 90°; Wrist over edge of table with forearm in full pronation
Stabilization:	Forearm stabilized to prevent pronation, supination, or shoulder rotation
Goniometer Axis:	Dorsal capitate
Proximal Arm:	Dorsal midline of the forearm
Distal Arm:	Parallel to longitudinal axis of the third metacarpal
Movement:	Frontal plane motion in medial and lateral directions
Expected ROM:	20° radial deviation and 30° ulnar deviation
Substitutions:	MCP abduction or adduction

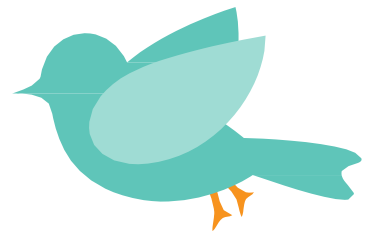




Hand Range of Motion

Motion	Recommend ed Testing Position	Stabilization	Axis	Proximal Arm	Distal Arm	Expected ROM	Start	End	End
MCP Flexion - Extension	Sitting with hand off table and forearm and wrist in neutral.	Stabilize metacarpals to prevent wrist motion. Do not hold other fingers in extension.	Over dorsal MCP joint	Dorsal midline of metacarpal	Dorsal midline of proximal phalanx	30-0-90°			
MCP Abduction- Adduction	Sitting with hand off table and forearm in full pronation and wrist in neutral	Stabilize wrist to prevent proximal motion	Over dorsal MCP joint	Dorsal midline of metacarpal	Dorsal midline of proximal phalanx	20-0-20°			
PIP Flexion- Extension	Sitting with hand off table and forearm, wrist, and MCP in neutral.	Stabilize proximal phalanx	Over dorsal PIP joint	Dorsal midline of proximal phalanx	Dorsal midline of middle phalanx	0-100°			
DIP Flexion- Extension	Sitting with hand off table and wrist in neutral	Stabilize middle phalanx	Over dorsal DIP joint	Dorsal midline of middle phalanx	Dorsal midline of distal phalanx	0-70°			
1 st CMC Palmar Flexion- Extension	Sitting with supinated on table and forearm and wrist in neutral	Stabilize carpal bones to prevent wrist motions	Palmar aspect of 1st CMC joint	Ventral midline of radius us- ing ventral surface of the radial head and styloid for reference	Ventral midline of 1st metacarpal	30-0-10°			
1 st CMC Palmar Abduction- Adduction	Sitting with ulnar border of hand on table and forearm, wrist in neutral	Stabilize carpal bones & second metacarpal to prevent wrist motions	Lateral aspect of radial styloid process	Lateral midline of 2nd meta- carpal. Reference center of 2 nd MCP joint	Lateral midline of 1st metacarpal Reference center of 1st MCP joint	70°			
Opposition		Stabilize 5th metacarpal to prevent wrist motions	Use ruler to measure distance between tip of thumb and 5 th finger						
1 st MCP Flexion	Sitting with ulnar border of hand on table and forearm, wrist in neutral	Stabilize 1st metacarpal to prevent wrist motion and flexion and position of thumb CMC	Dorsal aspect of MCP	Dorsal midline of metacarpal	Dorsal midline of proximal phalanx	0-50°			
1 st IP Flexion	Sitting with ulnar border of hand on table and forearm, wrist in neutral	Stabilize prox- imal phalanx to prevent flexion or ext- ension of MCP	Dorsal surface of IP	Dorsal aspect of proximal phalanx	Dorsal midline of distal phalanx	0-80°			





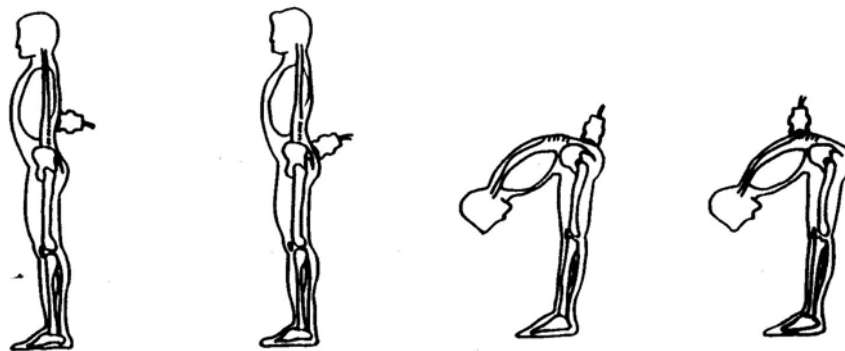
Spinal Inclinometry: Objective Measurement of Lumbar Spine Position and Range of Motion

Dual Inclinometer Technique

One inclinometer is placed over the sacrum and the other over T12-L1 spinous processes while the patient is standing in a relaxed erect posture. As with other techniques that require locating anatomical landmarks, the accuracy of the readings is markedly decreased with obese patients.

Angular readings are taken while holding the inclinometers in place while the patient is in the upright relaxed position. The patient is then instructed to maximally flex forward while the new angular readings are taken in the fully flexed position. The patient is instructed to continue forward bending until the pelvic motion is within 20% of the straight leg measurement used to determine hamstring flexibility.

The upper inclinometer reading represents gross motion while the lower inclinometer measures pelvic or hip motion. True lumbar motion is represented by the difference between these two measurements.



Single Inclinometer Technique:

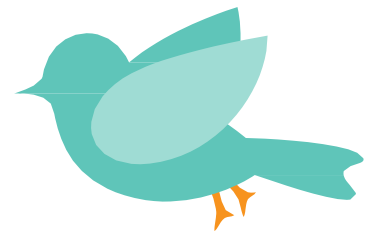
Method 1: The same technique as above is utilized except that the inclinometry measurements must be made at each site independently.

Method 2: The examiner's hands span the anterior and posterior parts of the iliac crest to allow measurement of pelvic mobility. The index finger of one hand is placed parallel to the floor along the top of the iliac crest. The opposite hand is holding the inclinometer at the T12-L1 interspace. The patient performs the same forward bending motion after the initial reading. Total motion is recorded from the inclinometer. The inclinometer is then placed across the plane connecting the thumb and index finger to determine pelvic motion. Calculation of pelvic and lumbar contribution to total motion is then calculated in the same way as with the dual inclinometer technique.

Extension range of motion is then measured in the same way as described above except the patient's motion is in the direction of extension.

Source: Mayer, et al. Spine 9(6), 1984. This method was shown to be reliable and have no statistical difference from radiographic determination of lumbar range of motion.

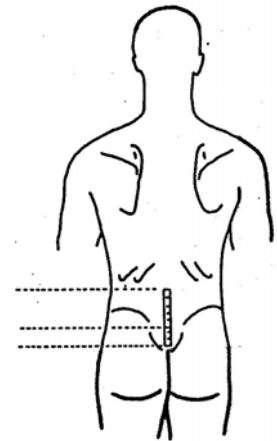




Linear Assessment of Lumbar Motion with Tape Measure

Vertebral Distraction (flexion) and Attraction (extension)

Anatomical landmarks (spinous processes) are identified and marked. A tape measure measurement is made of the distance between the two points. The patient is asked to flex or extend the spine and the new distance between the two points is remeasured. With flexion, the two points will be further apart, conversely, with extension the two points will approximate. The difference between the first and second measurement is an objective assessment of segmental or regional spine mobility between the initial anatomical landmarks.



Sources:

Beattie, et al. *Phys Ther* 67(3), 1987. Attraction tape measurement method demonstrated test-retest reliability and it differentiated normals from those with "significantly limiting low back pain". Intratester ICC = .95; Intertester ICC = .94

Balogun, et al. *J Ortho Sports Phys Ther.* 10(7), 1989. Compared gravity referenced goniometry to tape measurements of cervical range of motion. Both methods showed poor reliability with cervical flexion and good reliability with all other motions. Authors concluded that tape measurement was preferable because of its lower cost.

Tape measure mean normals:

Flexion	4.3 cm	(R) LF	12.9 cm	(R) Rot	11 cm
Extension	18.5 cm	(L) LF	12.8 cm	(L) Rot	11 cm

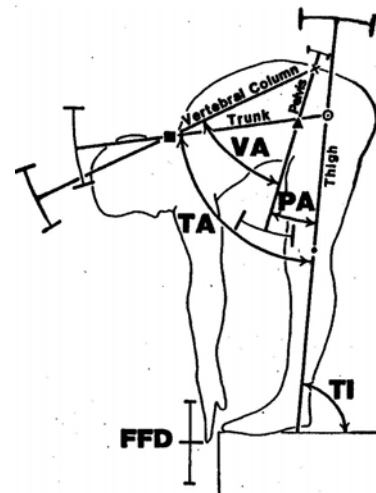
Chany-Yu, et al. *J Ortho Sports Phys Ther* 8(2), 1986. Also showed tape measure cervical range of motions to have a high intratester reliability for all motions (ICC = .80-.95).

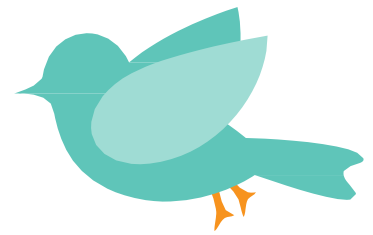
Toe Touch Test

The patient stands on a platform and maximally bends forward towards the toes with the knees extended. A measurement is taken from the fingertips to the platform surface. This value can be positive or negative. The value is positive when the patient can reach beyond the platform surface.

Source:

Kippers, et al. *Phys Ther.* 67(11), 1987. Studied the reliability of fingertip to floor distance and found excellent intertester reliability (ICC = .97). However, the reliability of vertebral flexion was very poor (ICC = .10). The authors concluded that at maximum trunk flexion, fingertip to floor distance is primarily a measure of hamstring flexibility. They suggest alternative methods for assessing lumbar mobility.

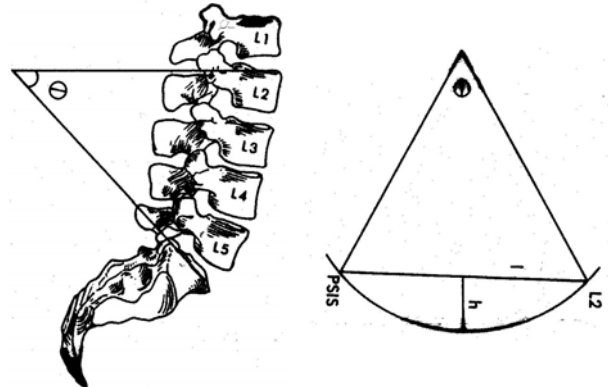




Assessment of Lumbar Lordosis

Flexible Ruler

A flexible cylindrical material is molded to the lumbar spine with markings at the L1-L2 and L5-S1 interspaces. The flexible ruler is then traced onto a piece of paper to determine the angle of the flexible ruler that in theory would be equal to the actual lumbosacral angle.



Source:

Hart, et al. *J Ortho Sports Phys Ther.* 8(4), 1986. Found good intertester reliability with this method (ICC = .97). They also established validity ($r = .87$) when compared to radiographic studies in a small sample ($n = 6$).

Bryan, et al. *J Ortho Sports Phys Ther.* 11(1), 1989. Found the flexible ruler has questionable clinical value in the assessment of lumbar lordosis or the actual lumbosacral angle. Average angles when measured between L1-2 and L5-S1 were 42° and 67° when measured from superior border of L1 and superior border of S1.

Lovell, et al. *Phys Ther.* 69(2), 1989. Reliability Study of Flexible Ruler

	LBP (ICC)	Normals (ICC)
Intratester Reliability – Examiner 1	.84	.94
Intratester Reliability – Examiner 2	.73	.83
Intertester Reliability	.43	.50

Rheault, et al. *J Ortho Sports Phys Ther.* 10(7), 1989. Found good intertester reliability using the spinocurve flexible ruler for measuring cervical range of motion. They used the occiput and C7 as anatomical landmarks. Intertester reliability in neutral was .80 and .90 in full cervical flexion. No attempt was made to establish validity.

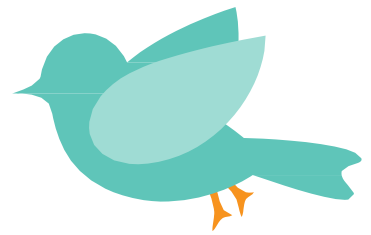
Visual Assessment of Lordosis:

Visual or photographic assessment of sagittal plane posture.

Source:

Bryan, et al. *J Ortho Sports Phys Ther.* 12(1), 1990. This study showed a 9% accuracy in ranking the amount of lumbar lordosis in clothed subjects. The strongest trend in intertester reliability was the rater's perception of increased lordosis might have been influenced by the gluteal prominence.





Hip Flexion:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with hips and knees in neutral rotation
Stabilization:	Trunk stabilized by body position
Goniometer Axis:	Femoral Greater Trochanter
Proximal Arm:	Parallel to midaxillary line of the trunk
Distal Arm:	Parallel to longitudinal axis of the femur in line with lateral femoral condyle
Movement:	Hip flexion with knee flexion allowed
Expected ROM:	120°
Substitutions:	Lumbar Spine flexion

Hip Extension:

Measurement Tool:	Universal Goniometer
Testing Position:	Prone with hips & knees in neutral and feet extending off end of the table
Stabilization:	Pelvis is stabilized through straps or manual fixation
Goniometer Axis:	Greater Trochanter
Proximal Arm:	Parallel to midaxillary line of the trunk
Distal Arm:	Parallel to longitudinal axis of femur in line with lateral femoral condyle
Movement:	Hip extension with knee extended. ASIS must remain on table.
Expected ROM:	30°
Substitutions:	Lumbar spine extension

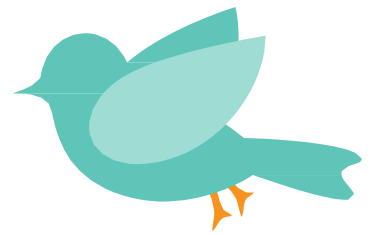
Hip Abduction:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with hips and knees in neutral and pelvis level
Stabilization:	Opposite innominate
Goniometer Axis:	ASIS on measured side
Proximal Arm:	Along a line between the two anterior superior iliac spines
Distal Arm:	Parallel to the long axis of the femur
Movement:	Abduction until motion is detected at the opposite anterior superior iliac spine
Expected ROM:	45°
Substitutions:	Hip external rotation, knee flexion/internal rotation, or lateral pelvic tilt

Hip Adduction:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with the opposite extremity abducted
Stabilization:	prn
Goniometer Axis:	ASIS on measured side
Proximal Arm:	Along a line between the two anterior superior iliac spines
Distal Arm:	Parallel to the long axis of the femur
Movement:	Adduction
Expected ROM:	30°
Substitutions:	Hip internal rotation or lateral pelvic tilt





Hip Internal and External Rotation:

Measurement Tool:	Universal Goniometer
Testing Position:	Sitting with the hip and knee flexed 90°. Opposite extremity abducted and resting on a foot stool
Stabilization:	prevent thigh abduction/adduction
Goniometer Axis:	mid-patella
Proximal Arm:	Perpendicular to the floor
Distal Arm:	Parallel to long axis of the tibia
Movement:	internal and external ROM
Expected ROM:	45° internal and external ROM
Substitutions:	thigh abduction/adduction

Alternate Test Position for Hip Internal/External Rotation:

Test Position:	prone with knees flexed 90°
Stabilization:	strap or manual fixation of pelvis
Diagnostic Findings:	similar measurements with hips flexed or extended indicate structural limitations while different amounts of motion in each position may indicate soft tissue tightness of the hip flexors/extensors

Knee Flexion:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine or reclined with hip and knee in neutral rotation
Stabilization:	Trunk and pelvis stabilized by body weight and position
Goniometer Axis:	Lateral epicondyle of the femur
Proximal Arm:	Parallel to the long axis of the femur & pointing at the greater trochanter
Distal Arm:	Parallel to the long axis of the fibula and pointing at the lateral malleolus
Movement:	The hip and knee are flexed as the heel moves toward the buttock
Expected ROM:	135°
Alternate Position:	Prone lying with the femur stabilized. Knee flexion motion may be decreased as the rectus femoris is now stretched over two joints. Prevent substitute motion of hip abduction and/or hip flexion.

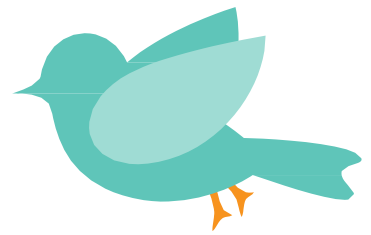
Knee Extension:

Measurement Tool:	Universal Goniometer
Testing Position:	Supine with hips and knees in neutral rotation; distal leg on bolster
Stabilization:	Trunk and pelvis stabilized by body weight and position
Goniometer Axis:	Lateral Epicondyle of the femur
Proximal Arm:	Parallel to the long axis of the femur & pointing at the greater trochanter
Distal Arm:	Parallel to the long axis of the fibula and pointing at the lateral malleolus
Movement:	Knee extension
Expected ROM:	0°. Hyperextension may be present up to 10-15°
Alternate Position:	Prone lying heel height technique

Tibial Internal/External Rotation:

Measurement Tool:	Universal Goniometer
Testing Position:	Prone lying with knee flexed 90 degrees
Stabilization:	Therapist manually stabilizes the femur
Goniometer Axis:	Parallel to the long axis of the tibia
Stationary Arm:	Parallel to the long axis of the tibia
Moving Arm:	Parallel to and in line with the bisection of the heel and the 2nd metatarsal shaft
Movement:	Internal and external tibial rotation
Expected ROM:	30° internal and 40° external range of motion. It is difficult to establish neutral rotation and it may be more test-retest reliable to record the total ROM





Talocrural Dorsiflexion:

Measurement Tool:	Universal Goniometer
Testing Position:	Prone with ankle off edge of table and knee extended
Stabilization:	Manual stabilization of tibia against the supporting surface
Goniometer Axis:	Lateral calcaneus at bisection of fibula and 5th metatarsal
Proximal Arm:	Parallel to the long axis of the fibula and pointing towards the fibular head
Distal Arm:	Parallel to the long axis of the 5th metatarsal
Movement:	The ankle is actively dorsiflexed
Expected ROM:	10° with knee extended; increased to 20 with the knee flexed
Substitutions:	Therapist must monitor for subtalar pronation. Watching for calcaneal eversion can monitor this.
Alternate Test:	The above procedure should be repeated with the knee flexed 90° to isolate soleus flexibility. A 10° increase in ROM to 20° should be expected

Talocrural Plantarflexion:

Measurement Tool:	Universal Goniometer
Testing Position:	Prone or supine with the knee in slight flexion
Stabilization:	Therapist stabilizes the lower leg
Goniometer Axis:	Lateral calcaneus at bisection of fibula and 5th metatarsal
Proximal Arm:	Parallel to the long axis of the fibula and pointing towards the fibular head
Distal Arm:	Parallel to the long axis of the 5th metatarsal
Movement:	The ankle is actively plantarflexed
Expected ROM:	30-50 degrees.
Alternate Position:	Prone lying

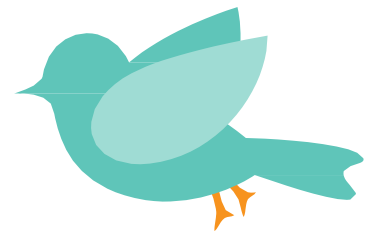
Subtalar Inversion/Eversion:

Measurement Tool:	Universal Goniometer
Testing Position:	Standing with bilateral stance
Stabilization:	No stabilization necessary
Goniometer Axis:	Just proximal to the achilles insertion on the calcaneus
Proximal Arm:	Parallel to the distal bisection of the lower leg
Distal Arm:	Parallel to the bisection of the calcaneus
Movement:	Patient can perform trunk and/or tibial rotation to maximally invert and evert the calcaneus
Expected ROM:	Approximately 10° of calcaneal eversion (pronation) and 20° of calcaneal inversion (supination) from the subtalar neutral position
Alternate Position:	Prone lying or unilateral stance positions. Maximal motion and greatest test-retest reliability has been demonstrated in the bilateral stance position

Metatarsophalangeal and Interphalangeal Flexion/Extension:

Measurement Tool:	Universal or Digit Goniometer
Testing Position:	Ankle in resting position of mild plantarflexion and the STJ is in neutral.
Stabilization:	Careful stabilization of metatarsals without metatarsal compression
Goniometer Axis:	Over the dorsum of the measured joint. If measuring the 1st or 5th the axis can be placed over the medial or lateral aspect of the joints, respectively.
Proximal Axis:	Parallel to the longitudinal axis of the metatarsal of the digit being measured
Distal Axis:	Parallel to the longitudinal axis of the metatarsal of the digit being measured
Movement:	Active flexion and/or extension of the distal digit being measured
Expected ROM:	1st MTP dorsiflexion - 20° passively and 60-70° actively; 1st MTP plantarflexion - 45° 2-5 MTP - 40° plantar/dorsiflexion; 2-5 PIP flexion - 35°; 2-5 DIP flexion - 60°
Substitutions:	The long toe flexors and extensors will effect ROM according to the positioning of the more proximal joints that they cross.





Subtalar and Forefoot Position and Range of Motion:

Y *Elvaru, 1988* - STJ ROM position and ROM Reliability: Prone NWB passive assessment

STJ Neutral Position:	Intratester Reliability	= .77
	Intertester Reliability	= .25
STJ ROM:	Intratester Reliability of Inversion	= .62
	Intratester Reliability of Eversion	= .59
	Intertester Reliability of Inversion	= .15 - .32
	Intertester Reliability of Eversion	= .12 - .17
	(reliability decreased with referencing)	

** *Elvaru, 1988* studied prone passive df/pf ROM reliability

plantarflexion intratester reliability	= .86	plantarflexion intertester reliability	= .72
dorsiflexion intratester reliability	= .97	dorsiflexion intertester reliability	=

.50 *Smith-Orricchio, 1990* - found similar poor reliability for prone assessment of STJ neutral and ROM - .25 ever, .42inv, .60 STJN interrater reliability

Functional Reach Protocol

Functional Linear Methods – like Apley's scratch and prone heel height

Goniometric Reliability of Shoulder Measurements in a Clinical Setting. *Phys Ther* 67:668-673, 1987
Reliability of Functional and Goniometric Measurements in the Assessment of Shoulder Range of Motion. Mulligan, et al, 1998

Intratester and Intertester Reliability of Shoulder Goniometric Measures

MOTION	Intratester ICC	Intertester ICC
Flexion	.98/.90	.89/.75
Extension	.94	.27
Abduction	.98	.87
Horz Abduction	.90	.30
Horz Adduction	.95	.41
External Rotation	.99	.88
Internal Rotation	.94/.94	.55/.90

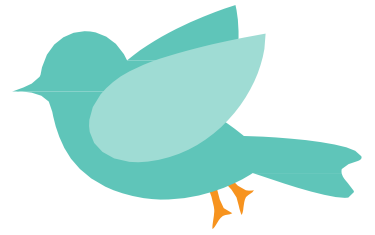
Effect of Scapular Stabilization on Rotation ROM

	Women	Men
% decrease in IR	29%	18%
% decrease in ER	2%	12%

Y

Y *Smith-Orricchio, 1990* - found higher interrater reliability of STJ neutral in standing .75 unilateral stance; .91 bilateral stance





- Y *Picciano, 1995* - found poor intra and intertester reliability in assessing STJN in OKC, CKC, and navicular drop test assessments with inexperienced testers. SEM was approximately 2.5° for both goniometric methods and 2-2.5 mm for navicular drop test.
- Y *Lattanza, 1988* - found significantly more eversion ROM in standing weight bearing assessment. Average STJN position of 4.2° varus. NWB calcaneal eversion from STJN was 6.5° and 10.4° in WB, a 37% increase.

Shoulder External Rotation in Dependent Position (20-30° Abduction) Intratester ICC = .88-.93

Shoulder External Rotation in Dependent Position (20-30° Abduction) Intertester ICC = .80-.85

MacDermid, et al. J Hand Therapy 12:187-192, 1999.



SELF NOTES

